

OFFICIAL JEE MAINS & ADVANCED SYLLABUS

Complete Curricular Outline (Class 11 & Class 12 Coursework)

1. Physics Syllabus Core Topics

Class 11 Physics Topics: Foundation of classical mechanics, gravity dynamics, kinetic theories, thermodynamic cycles, and wave mechanics.

- Units and Dimensions: Dimensional Analysis, Least count, Vernier Callipers, Screw Gauge errors.
- Kinematics: Projectiles, Frame of Reference, Relative Velocity, Uniform/Non-uniform motion.
- Laws of Motion: Newton's Laws, Friction coefficients, Centripetal & Coriolis pseudo forces.
- Work, Energy and Power: Work-Energy Theorem, Conservative forces, Elastic collisions.
- Rotational Motion: Angular momentum, Moment of Inertia, Rolling and Torque dynamics.
- Gravitation: Kepler's Laws, Escaping Velocity, Satellite orbits potentials.
- Properties of Solids and Liquids: Young's Modulus, Surface Tension, Viscosity, Bernoulli theorem.
- Thermodynamics: Heat engine Carnot efficiency, Second Law, isothermal/adiabatic expansions.
- Kinetic Theory of Gases: Ideal gas assumptions, Mean free path, Degree of Freedom values.
- Oscillations and Waves: SHM dampings, Wave Speed string mediums, Acoustics Doppler shifts.

Class 12 Physics Topics: Electromagnetism, light wave physics, semiconductor devices, and atomic modern physics models.

- Electrostatics: Coulomb's law, Gauss theorem flux, electrostatic potential capacity.
- Current Electricity: Ohm's law, Kirchhoff's junction rules, Potentiometer, Wheatstone bridge.
- Magnetic Effects of Current & Magnetism: Biot-Savart, Ampere's law, Cyclotron, Magnetic fields.
- Electromagnetic Induction: Faraday's law, Lenz's law, Self & Mutual Inductance calculations.
- Alternating Current: LCR circuit resonance, Quality factor, Power in AC, Transformer ratios.
- Electromagnetic Waves: Displacement current, EM wave spectrum propagation.
- Optics: Reflection, Refraction, Lens formulas, Wave interference Young double slit, Diffraction.
- Dual Nature of Matter & Radiation: Photoelectric effect, de Broglie wave relation.
- Atoms and Nuclei: Rutherford, Bohr hydrogen model, Radioactivity half-lives, Nuclear fusion/fission.
- Electronic Devices: Semiconductor p-n junction diode, I-V characteristics, Logic Gates.
- Experimental Skills: Galvanometer resistance, Focal length lenses, Speed of sound resonance tube.

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Part 2: Chemistry and Mathematics Curriculum Outline

2. Chemistry Syllabus Core Topics

Physical Chemistry: Basic stoichiometry, atomic quantum structures, thermodynamic state systems, and chemical/electro-kinetics.

- Some Basic Concepts of Chemistry: Mole fraction, Molality, Molarity, Stoichiometry, Empirical formulas.
- Atomic Structure: Bohr model, Quantum numbers, Aufbau principle, Hund's rule, dual nature.
- Chemical Bonding: Lewis structure, VSEPR model, Hybridization, Molecular orbital configurations.
- Chemical Thermodynamics: First/Second laws, Enthalpy, Entropy, Gibbs energy spontaneity.
- Solutions: Raoult's law, Colligative properties (elevation of boiling point, osmotic pressure).
- Equilibrium: Law of mass action, Le Chatelier, pH, solubility product buffer solutions.
- Redox & Electrochemistry: Nernst equation cell EMF, Conductance, electrolysis Faraday rules.
- Chemical Kinetics: Collision theory, first order rates, Arrhenius equation activations.

Inorganic & Organic Chemistry: Periodic coordination ligands, salt analysis, reaction mechanisms, and bio-synthesis.

- Classification of Elements: Periodic properties, Atomic radius, Electronegativity, Ionization enthalpy.
- p-Block Elements: Group 13 to 18 elements trends, oxides structures, anomalous properties.
- d- and f-Block Elements: Transition series configuration, Lanthanoids contraction, interstitial compounds.
- Coordination Compounds: Werner theory, IUPAC rules, Isomerism, VBT, Crystal Field Theory.
- Salt Analysis: Identification of acidic/basic radicals (Pb^{2+} , Cu^{2+} , Fe^{3+} , Cl^- , SO_4^{2-}).
- Metallurgy: Concentration, reduction, refining operations of Iron, Copper, Zinc.
- Organic Chemistry: Hydrocarbons, Grignard RMgX , Aldol reactions, Halides, Alcohols, Phenols, Ethers.

3. Mathematics Syllabus Core Topics

Algebra, Calculus, and Geometry: Functional analysis, matrix calculations, limits, differential systems, vectors, and probability.

- Sets, Relations and Functions: Types of relations, equivalence relations, domain, range.
- Complex Numbers & Quadratic: Argand plane, Euler formula, roots coefficients relations.
- Matrices & Determinants: Inverse matrix, Adjoint, Cramer's system rules linear equations.
- Permutations and Combinations: Circular permutations, binomial combinations notation.
- Sequences & Series: Arithmetic, Geometric progressions, AGP, summation of series.
- Calculus (Limits, Integrals, Diff. Eq.): L'Hospital rules, derivatives, definite integral areas.
- Co-ordinate Geometry: Straight lines, circles, parabola, ellipse, hyperbola standard equations.
- Vector Algebra & 3D Geometry: Dot/cross products, lines/planes projections, shortest distances.
- Probability & Statistics: Bayes theorem, Poisson/Binomial variables, mean/variance deviations.